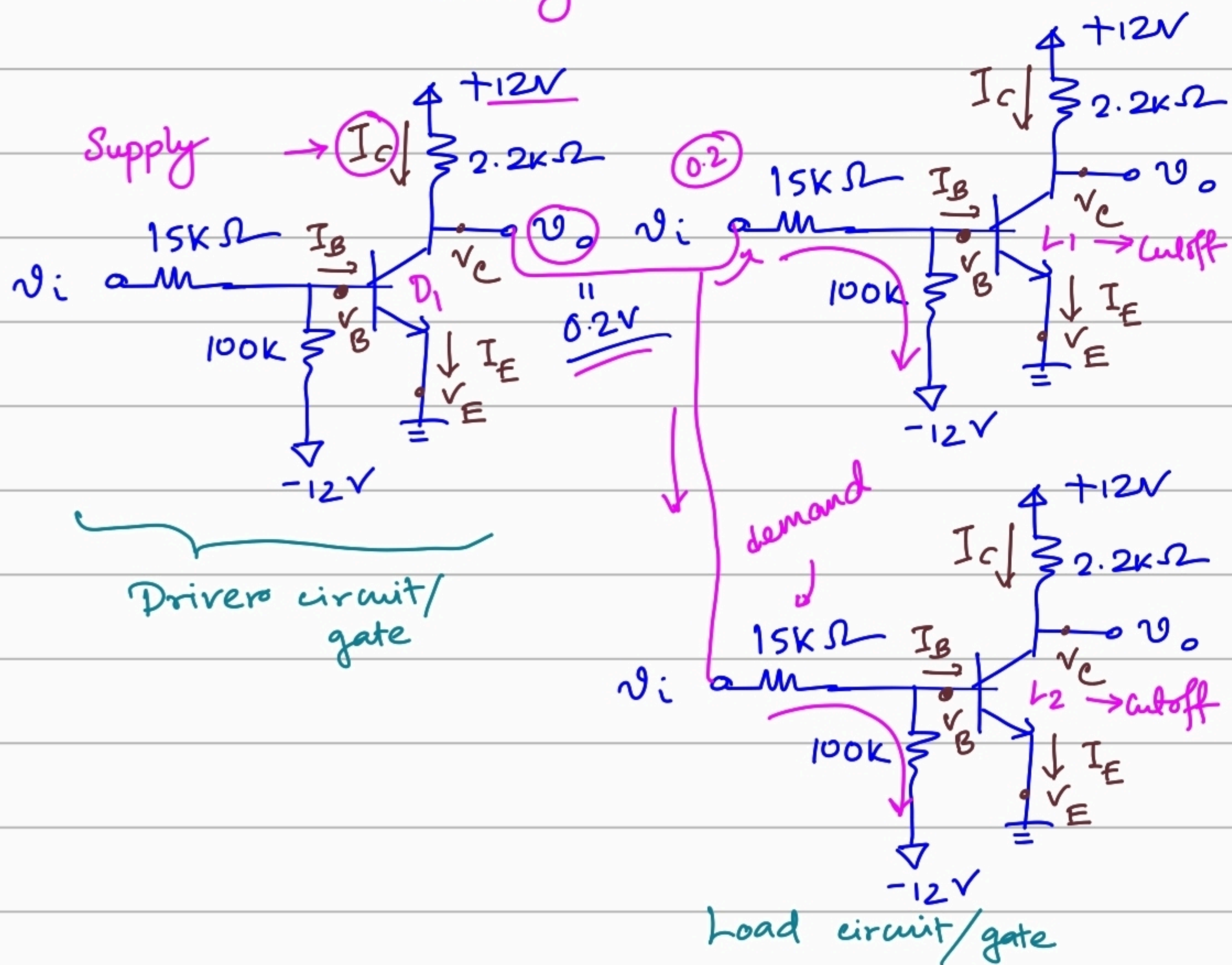


Fanout: The number of input terminals are connected to a output terminal.

Maximum fanout (fanout): The maximum number of logic inputs of the logic family that an output terminal can drive, is called maximum fanout.

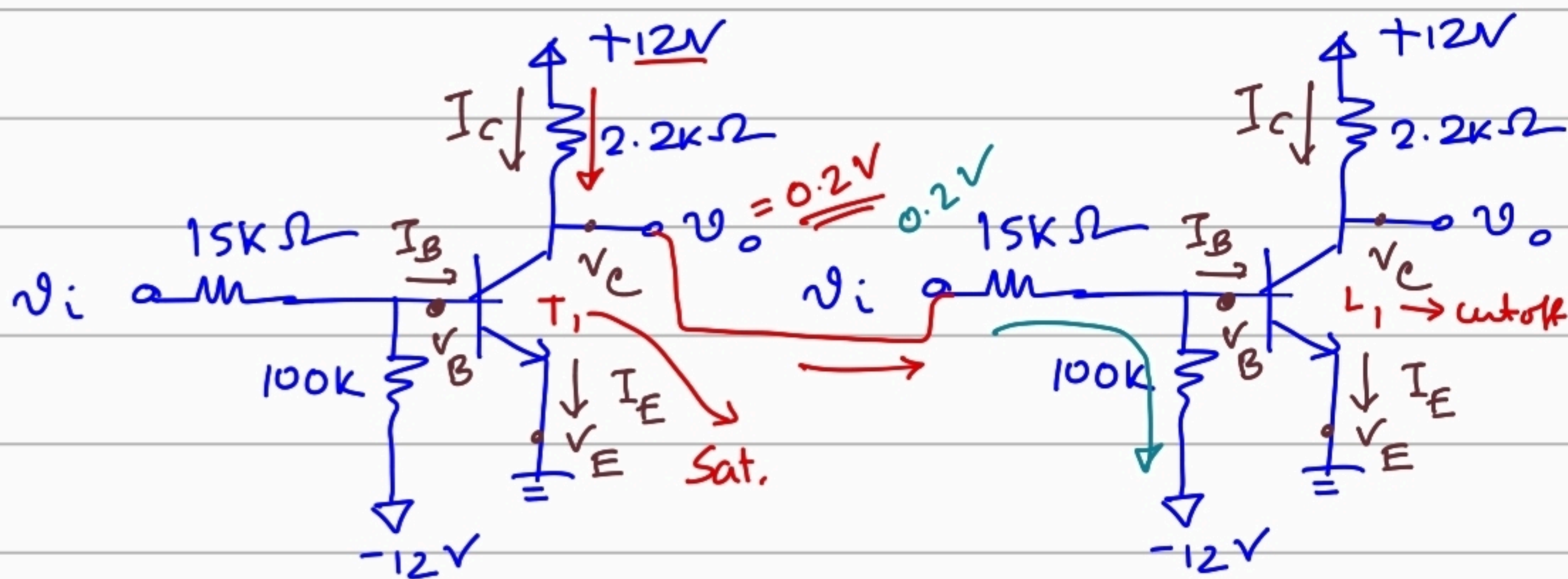


Question: Calculate the maximum fanout of the above RTL circuit. Assume $V_{OH} = 10V$.

Case ①: The output of the drivers circuit is low.

* Maximum supply current :- The maximum of current can be delivered by the driver circuit without changing its operating mode.

* Individual demand load current :- The maximum amount of current needed by each individual load while driven by driver circuit.

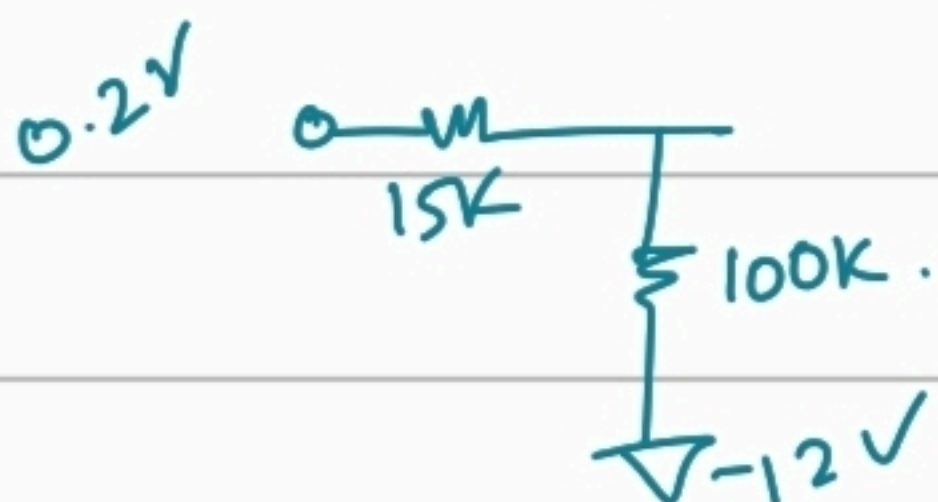


Driver circuit $T_1 \rightarrow$ Saturation. $V_o = 0.2V$

Load circuits $L_1, L_2 \rightarrow$ cutoff. $I_B = 0$.

* Maximum supply current, $\frac{12-0.2}{2.2k} = 5.3636 \text{ mA}$

* Individual demand load current, $\frac{0.2-(-12)}{(15+100)k} = 0.106 \text{ mA}$



maximum fanout for this case,

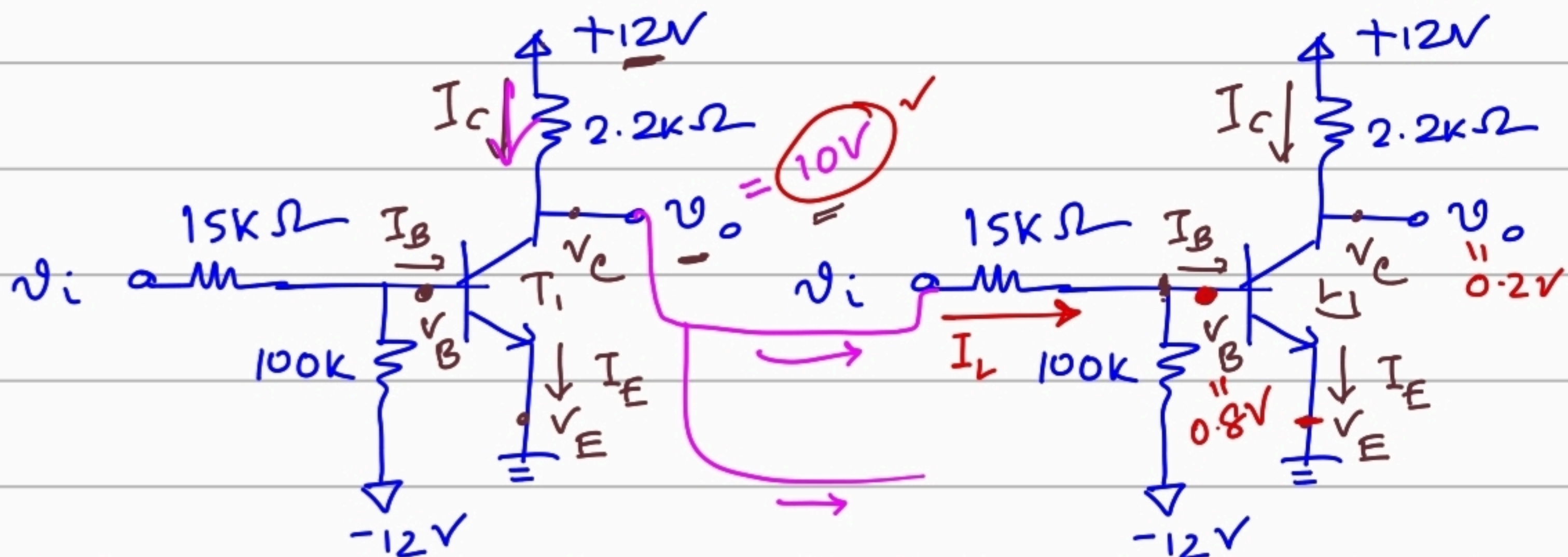
$$= \left\lfloor \frac{5.3636}{0.106} \right\rfloor = \left\lfloor 50.6 \right\rfloor = 50$$

* If we connect 51 load circuits, then demand exceeds supply. Thus driver circuit will malfunction.

Case ② : The output voltage of the driver circuit is high.

* If we connect load circuits when output of driver is high, the output voltage of driver starts to drop.

$$V_{OH} = 10V$$



* Driver transistor, $T_1 \rightarrow$ cutoff, load circuit's transistors T_2

* Maximum supply current = $\frac{12-10}{2.2k\Omega} = 0.909mA$ Sat.

* Individual load current, $I_L = \frac{10-0.8}{15k} = 0.6133mA$

* Maximum fanout for this case = $\left\lfloor \frac{0.909}{0.6133} \right\rfloor = \lfloor 1.48 \rfloor = 1$

▢ Overall maximum fanout must be calculated by considering worst case scenario. To prevent malfunction of driver circuit, we need to take the minimum of the two cases.

▢ Maximum fanout, $= \min(50, 1) = 1$